

Summary

Research fields: Human-Robot Interaction、Human-Computer Interaction、User Experience Design/Research、Human-Centered Physical AI

I am a Human-Robot Interaction (HRI) researcher with an engineering background and hands-on experience in robot interface development, teleoperation system integration, field testing, and user behavior analysis. My work bridges system development and empirical research to understand the challenges that arise when robots move into real-world social environments (e.g., service, medical, industry), including user confusion, lack of trust in the robot, barriers to acceptance or unclear interaction authority. With a multidisciplinary background, I aim to contribute to the application of robots and AI through both teaching and research.

我是人與機器人互動 (HRI) 的研究者，擁有機器人操作介面開發、遠端操作系統整合、真實場域測試與使用者行為分析的經驗。我的工作結合系統開發與實證研究，旨在理解機器人進入真實社交環境時(例如服務、醫療、工業用途)所面臨的挑戰，包括使用者對於機器人感到困惑、對機器人的信任不足、對機器人的接受度或是否能辨識機器人在互動中的權威性等問題。我目前正在尋找大學教職，期望能以教學及研究為機器人與人工智慧的應用做出貢獻。

Education

2019-2025 Ph.D, Dept of Informatics, Social Informatics, Kyoto University 日本京都大學資訊研究所博士

Advisor: Professor Takayuki Kanda 神田崇行

Main research interest: Human-Robot Interaction (HRI), Human-Computer Interaction (HCI), moral interaction in robot and avatar robot

My thesis proposes the concept of authoritative presence, an impression that elicits respect-based, internally perceived authority and thereby encourages people to cooperate with a robot. Unlike traditional approaches that rely on coercion or externally imposed authority—which may be difficult to sustain and can evoke negative emotional responses, authoritative presence operates through users' perceptions and experiences during interaction. We posit that authoritative presence can produce outcomes similar to those of formal authority, but through internally motivated cooperation rather than compliance driven by external force or obligation. Through three empirical studies, we examined how a robot's behavior and the operator's embodiment contribute to the formation of authoritative presence.

我的博士論文提出「權威性臨場感」(authoritative presence) 的概念，指的是一種能引發尊重感、並使使用者在互動中感受到內在權威性的印象，進而促使人們願意與機器人合作。不同於傳統上依賴強制力或外部賦予權威的方式，這類權威不一定容易長期維持，也可能引發負面情緒；權威性臨場感則是透過使用者在互動過程中的感知與經驗而產生。我們主張，權威性臨場感能帶來類似正式權威的效果，但其作用機制並非來自外在壓力、義務或強迫性的服從，而是來自使用者內在動機所形成的合作意願。本論文透過三項實證研究，探討機器人的行為設計與操作者的具身呈現如何共同影響權威性臨場感的形成。

2014-2017 M.S. Institute of Information System and Applications, National Tsing Hua University

國立清華大學資訊系統與應用研究所碩士

2009-2013 B.S., Computer Science, National Tsing Hua University

國立清華大學資訊工程系學士

Employment

- 2025.11- Postdoctoral Researcher, Dept of Information Management, National Taiwan University,
國立台灣大學博士後研究員
- 2018-2019 Research Assistant, National Taiwan University, IOX Center 國立台灣大學 IOX 中心研究助理
- 2017-2018 QA Software Engineer, HTC, DeepQ 宏達國際電子軟體工程師
- Did research and designed testing automation workflow and system with Angular, Python or any other needed programming languages.
 - Cooperated with UX team to do user research and improved testing flow.
- 2013-2014 Software engineer, Otto Group, OSP Taipei 甌圖軟體開發股份有限公司軟體工程師
- Built backend software of e-commerce system, such as staff-end software, order and invoice processing, warehouse managing with Ruby on Rails, Qt, PL/SQL, Shell Script.
 - Cooperated with oversea colleges from Germany, Japan and Russia.

Publications

- [J2] **Chang, Y. C.**, Rea, D. J., & Kanda, T. (2026). Effects of Human Operator Embodiment on Approaching Stranger Behavior. ACM Transactions on Human-Robot Interaction. <https://doi.org/10.1145/3811029> (impact factor 7, Q1)
- [J1] **Chang, Y. C.**, Rea, D. J., Lee, C. J., Kanda, T., & Chen, B. Y. (2025). Designing Authoritative Presence in Social Robots. ACM Transactions on Human-Robot Interaction, 15(1), 1-30. <https://doi.org/10.1145/3748262> (impact factor 7, Q1)
- [C3] **Chang, Y. C.**, Rea, D. J., & Kanda, T. (2024, March). Investigating the impact of gender stereotypes in authority on avatar robots. In Proceedings of the 2024 ACM/IEEE International Conference on Human-Robot Interaction (pp. 106-115). <https://doi.org/10.1145/3610977.3634985> (25% acceptance rate)
- [C2] Tsai, H. R., **Chang, Y. C.**, Wei, T. Y., Tsao, C. A., Koo, X. C. Y., Wang, H. C., & Chen, B. Y. (2021, May). Guideband: Intuitive 3d multilevel force guidance on a wristband in virtual reality. In Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (pp. 1-13). <https://doi.org/10.1145/3411764.3445262> (24% acceptance rate)
- [C1] **Chang, Y. C.**, Wang, H. C., Chu, H. K., Lin, S. Y., & Wang, S. P. (2017, February). AlphaRead: Support unambiguous referencing in remote collaboration with readable object annotation. In Proceedings of the 2017 ACM Conference on Computer Supported Cooperative Work and Social Computing (pp. 2246-2259). <https://doi.org/10.1145/2998181.2998258> (35% acceptance rate)

Teaching Experience

- 2024 Teaching Assistant, Information System Analysis, Professor Takayuki Kanda, Kyoto University
- 2014-2016 Teaching Assistant, Software Studio, Professor Hao-Chuan Wang, NTHU

Academia Service

- Reviewer CSCW 2020, CHI 2020 late breaking works, HAI 2023, HAI 2024, RO-MAN 2025, HRI 2025, HRI 2026, THRI (Transactions on Human-Robot Interaction), International Journal of Social Robotics
- SV MobileHCI 2019

Awards and Honors

- 2020-2023 Taiwan Major Field Scholarship 教育部重點領域獎學金 (規則比照公費留考)
- 2016 National Tsing Hua University Internation Exchange Student Scholarship
國立清華大學交換學生獎學金
- 2009 College of Electrical Engineering and Computer Science Excellent Student Scholarship
國立清華大學電資學院優秀學生獎學金